# BiWrite User Guide

Version 0.2.2

October 2026

Source code https://github.com/zzccppp/biwrite
Writing skill research-builder https://github.com/qzkinhit/research-builder

**Abstract**

BiWrite is an editor for writing an academic paper in English and Chinese at once. You write on the left, the right pane shows the other language paragraph by paragraph, and only paragraphs you change are translated again. A LaTeX paper compiles into a PDF inside the app, and a click on a sentence in the PDF selects that sentence in the source. A writing assistant polishes, edits, answers questions, writes new paragraphs in the manner of a published paper and writes figures, with the rules of the research-builder writing skill. Section 2 follows one paper from the template to submission, and the sections after it go through each feature step by step. In the screenshots, red boxes mark the buttons to use and the areas to look at, and the numbers in the text, such as **(1)**, refer to the numbers next to the boxes.

## 1 Features at a glance

Each feature name below links to its section.

**The writing workflow**
Ten steps from a new paper on a template, through writing, imitating a published paper, figures, the PDF and editing the Chinese, to the Chinese version.

**Bilingual writing**
English on the left, the Chinese paragraph by paragraph on the right, and only changed paragraphs are translated again. The swap button in the toolbar lets you edit the Chinese instead, with the English following and unchanged paragraphs kept word for word. Swapping early, paragraphs not translated yet are filled in on the left as their translations arrive.

**A paper written in Chinese**
A Chinese file opens as the Chinese side and is translated into English, and saving always writes the file's own language. A document taken for the wrong language is set right in one step.

**LaTeX builds and PDF navigation**
The PDF compiles inside the app. A click on a sentence in the PDF selects it in the source, ready to polish. The paragraph at the cursor can be found in the PDF, and a build error opens its line.

**Chinese PDF**
Once the English is translated, the Chinese PDF compiles in one step, and the Chinese PDF and a stand-alone Chinese .tex can be exported.

**Your existing Chinese version**
`paper_zh.tex` or `sections_zh/` pair with the English paragraph by paragraph. Edit either side and the matching paragraph of the other file is updated in place, with the rest unchanged.

**Writing assistant**
Polish, edit by instruction, ask, write new paragraphs in the manner of a reference paper, write

figures, or rewrite a translation and let the paragraph follow. A reference paper can be added as its `.tex` file, and packages a figure needs are added on insert. Jobs run in the background, and you decide what is applied.

**Paper templates**
VLDB, ICLR 2027 and IEEE TII built in. Your own projects can be imported as templates, and the open project exported.

**APIs and key pools**
OpenAI-compatible, OpenAI Responses and Anthropic APIs. A provider holds several keys, uses them evenly and moves to another key on rate limits.

**Request log and updates**
The log records the model, reasoning effort and service tier of every request. Any version published on GitHub can be installed from inside the app.

**Close to the tray**
Closed, the window hides and BiWrite stays in the menu bar or notification area with the document and its translations. A click on the icon brings it back.

Figure 1 shows the main window: **(1)** is the source, **(2)** the translation.




Figure 1 The main window: (1) English LaTeX on the left, (2) the Chinese paragraph by paragraph on the right

## 2 The writing workflow

This section follows one paper from a new project to submission in ten steps. Each step gives the essentials and points to the section with the details.

### 2.1 Step 1: A new paper from a template

Click *New* in the toolbar, find the venue in *Papers and templates* (Figure 2 **(1)**), click *New paper…* **(2)** and name the folder of the paper. BiWrite copies the template and opens its main file. Section 4.7 shows how to add templates of your own.




Figure 2 (1) The ICLR 2027 template, (2) New paper

## 2.2 Step 2: Writing with the translation alongside

Write English on the left. 0.8 seconds after you stop typing, your changes reach the translation queue, and a changed paragraph is translated once it has stayed unchanged for 5 more seconds. The right pane then shows its Chinese (Figure 1). Settings can change the 5 seconds to anything from 0 to 60 (Section 10.6). A first draft in Chinese works too. When BiWrite finds the text is Chinese, it translates it into English instead (Section 3.4).

## 2.3 Step 3: Writing in the manner of a published paper

For paragraphs with a familiar shape, such as an introduction or related work, the writing assistant can write in the manner of a published paper.

1. Put the cursor in the paragraph the new text should follow, press ⌘J and choose *Write* (Figure 3 **(1)**). **(2)** shows the paragraph the new text follows. With the cursor on an empty line, the new text goes there.
2. Write what you want at **(3)**: the subject, how many paragraphs, how long. For example, *two paragraphs motivating in-context learning on graphs, written like the introduction of the reference paper*.
3. Click + *File* **(4)** in the *References* row and choose the paper to imitate, such as the `.tex` file of the UniClean paper. A LaTeX file gives its body without comments, and a long paper is cut after 120,000 characters. The chip shows the file name and its length **(5)**. + *Text* adds a pasted passage instead.
4. Click *Run* **(6)**.

The model follows how the reference paper arranges its paragraphs, its sentence patterns and how it argues, and takes the content from your instruction and the paper around the cursor. It does not copy the reference's sentences, claims, numbers or citations. A result your paper does not give yet appears as a marked placeholder such as `[result]` for you to fill in.

The card (Figure 4) shows the new paragraphs **(1)**, their Chinese paragraph by paragraph **(2)** and notes on how they were written **(3)**. *Insert* **(4)** puts them after the target paragraph, and the right pane shows the Chinese from the card without another request (Figure 5).

## 2.4 Step 4: Polish, edit and ask

*Polish* evens out the wording, *Edit* changes a paragraph as you say, and *Ask* checks an argument. Section 6 has the details.

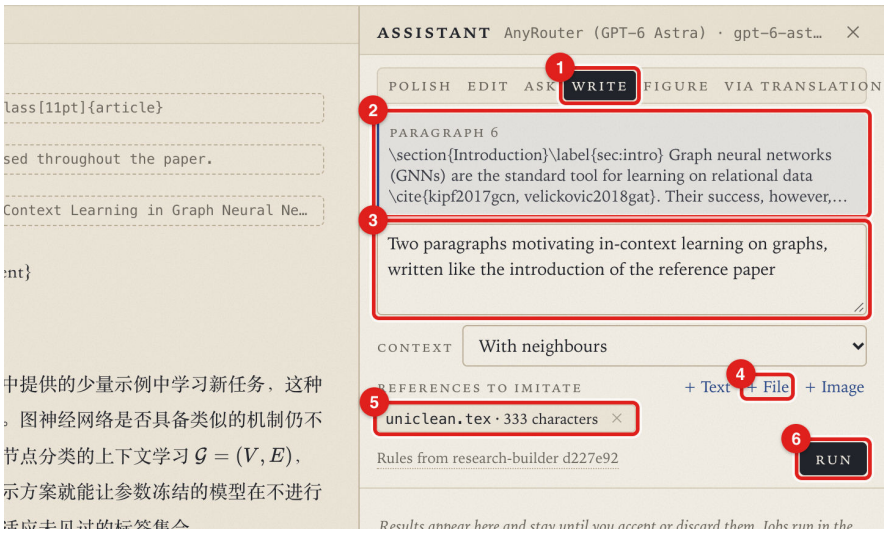



Figure 3 (1) Write, (2) where the new text goes, (3) what to write, (4) add a reference paper, (5) the reference added, (6) Run




Figure 4 (1) The new paragraphs, (2) their Chinese, (3) notes, (4) Insert

## 2.5 Step 5: Figures and tables

Choose *Figure* and describe the figure or table, for example *accuracy on the four benchmarks as a table*. The card gives LaTeX ready to insert (Figure 6 **(1)**). The request lists the packages the document loads and asks for code that uses them first. When the code needs a package the document does not load, the card names it **(2)**. If the open file is the main file, *Insert* **(3)** puts the code after the paragraph and adds the `\usepackage` lines right before `\begin{document}` (Figure 7). In a section file, the card names the lines to add to the main file's preamble.

To draw in the style of an existing figure, choose *Figure* (Figure 8 **(1)**), say what to draw and from which data **(2)**, add the figure with + *Image* **(3)** in the *References* row, which shows it as a thumbnail chip **(4)**, and click *Run* **(5)**. The model draws the same kind of chart with the same layout, colours, marks, line styles, fonts and legend placement, in TikZ or pgfplots, with the data and labels from your instruction and the paper, never the numbers in the image. Images can be PNG, JPEG, WebP

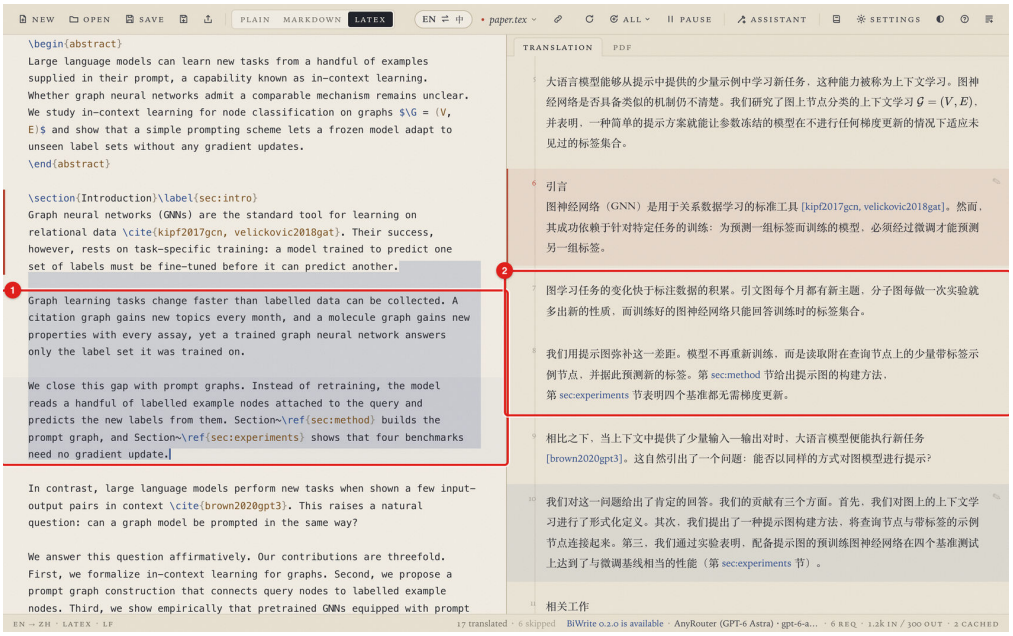



Figure 5 (1) The new paragraphs on the left, (2) their Chinese on the right




Figure 6 (1) The table, (2) the packages it needs, (3) Insert

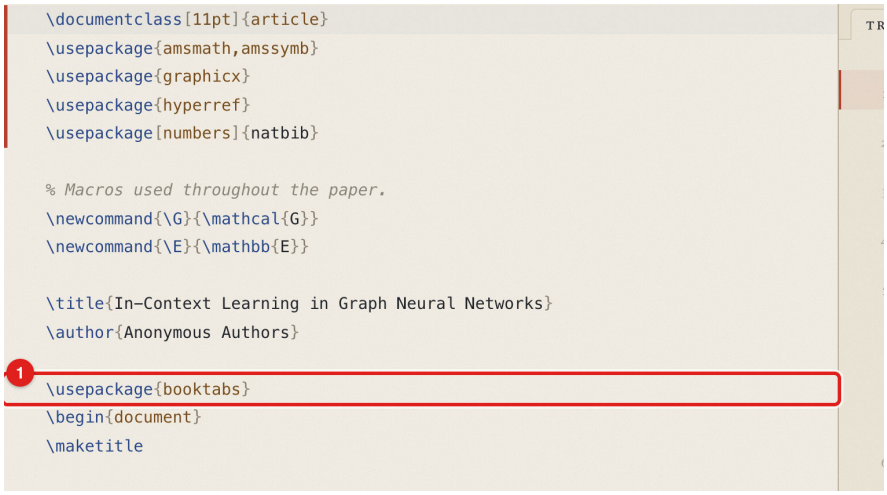



Figure 7 (1) The package added to the preamble on insert

or GIF of up to 8 MB each, and the model must accept images.




Figure 8 A figure in the style of a reference image: (1) Figure, (2) what to draw, (3) add the image, (4) the image added, (5) Run

## 2.6 Step 6: The PDF, and editing from it

Open the PDF tab and press ⌘B. A click on a sentence in the PDF selects it in the source, ready to polish or edit (Section 4). The line numbers in the margin of a submission do not get in the way.

## 2.7 Step 7: Editing the Chinese

Click *EN* ⇄ 中 to edit the Chinese on the left, with the English following (Section 3). To change one paragraph only, click the pencil on its Chinese on the right and rewrite the Chinese with *Via translation*. The English is revised to say the same (Section 5).

## 2.8 Step 8: The Chinese PDF and .tex

In the PDF tab, switch to *Chinese*, compile, and save the PDF and the Chinese `.tex` with *Save PDF as…* and *Export Chinese .tex…* (Section 4.5).

## 2.9 Step 9: Keeping the Chinese version in step

Put the exported Chinese `.tex` next to the English main file as `paper_zh.tex`. The next time the English file opens, the two pair up, and an edit on either side updates the matching paragraph of the other file in place (Section 5).

## 2.10 Step 10: Saving and keeping versions

⌘S saves, and *Save As* keeps the versions of each stage. Once the paper is done, *Export this project…* under *New* packs it as a template for the next paper.

# 3 Everyday writing

## 3.1 Open and edit

*Open* in the toolbar (Figure 9 **(1)**, ⌘O) opens a `.tex`, `.md` or `.txt` file. A LaTeX project of several files lists its files in the title bar **(2)**, to move between them. Each block on the right is one paragraph, heading or caption of the source. Click a block and the cursor moves to its paragraph, which is highlighted on both sides **(3)**.

Only the paragraphs you changed are translated again. Each is sent once it has stayed unchanged for the wait set in Settings (5 seconds by default), and its translation streams in as it arrives. A language swap, the PDF of the other language, an export and saving a paired file send the waiting paragraphs at once. Math, citations, references, labels, URLs and comments reach the model as placeholders and come back exactly as they were.

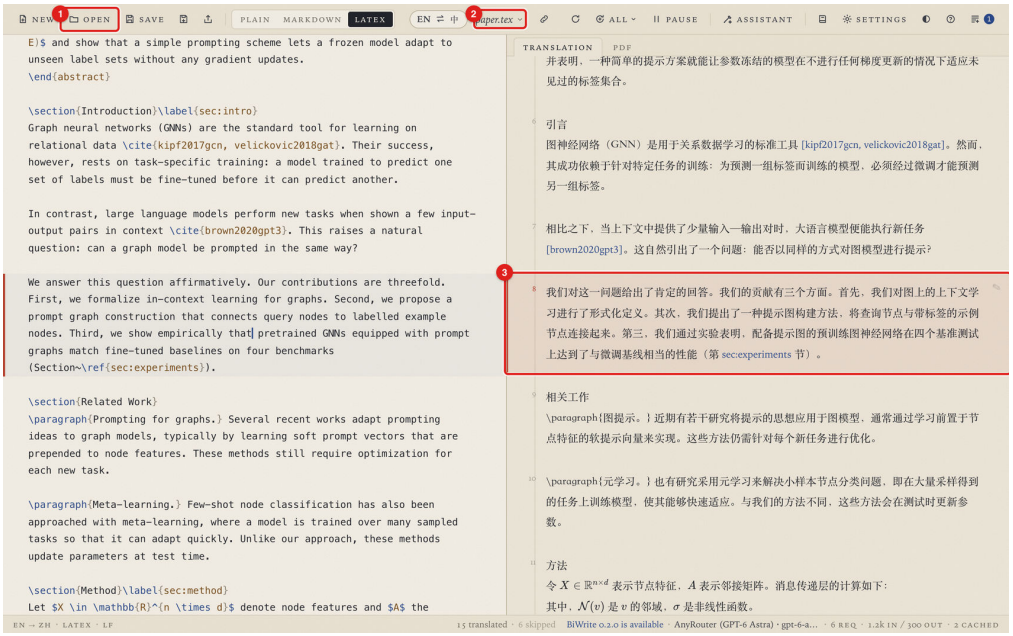



Figure 9 (1) Open a file, (2) the files of the project, (3) the paragraph's translation

## 3.2 Switching the edited language

The *EN* ⇄ 中 button in the middle of the toolbar (Figure 10 **(1)**) sets which language you edit on the left.

1. By default you edit English on the left and read Chinese on the right.
2. After the swap, the left pane holds the Chinese (Figure 11 **(2)**) and you edit it, with the English on the right **(3)**. Only the paragraphs you change get new English, revised from your previous English. Every other paragraph keeps your exact English.
3. Click again **(1)** to edit English. Paragraphs you did not change come back word for word.

A swap needs every paragraph translated. If some are still being translated, the button waits and swaps once they are done. A second click swaps at once. The paragraphs not translated yet keep their text for the moment, are translated in the background and are filled in on the left as their translations arrive (Figure 12). A paragraph you edit before that keeps your edit. When you swap back, their English comes back word for word. While translation is paused nothing is filled in, and *Continue with the rest* in the *All* menu does it.

Saving always writes the file's own language. While you edit the Chinese of an English file,

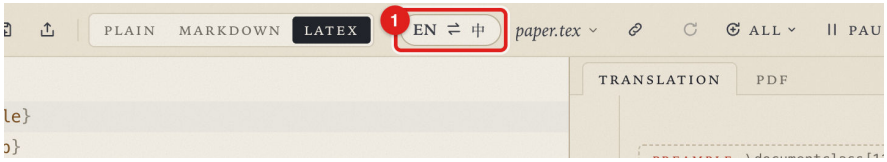



Figure 10 (1) The button that swaps the edited language

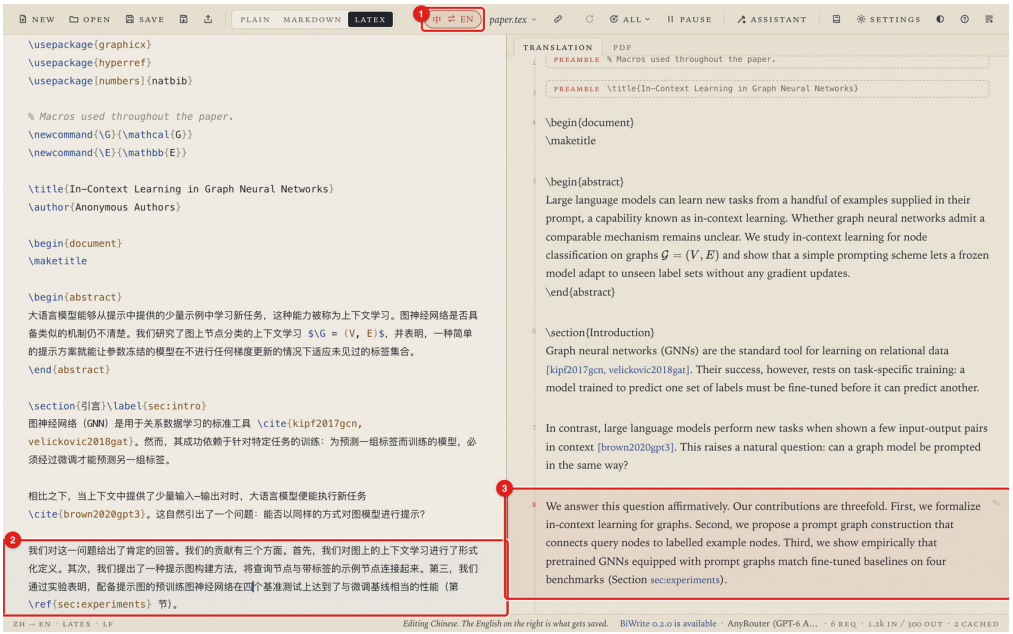



Figure 11 After the swap: (1) the button reads 中 ⇄ EN, (2) the Chinese is edited on the left, (3) the English follows on the right




Figure 12 Swapping early: (1) editing the Chinese, (2) a paragraph still being translated, (3) the note in the status bar

saving writes the English composed from the translations. With a paired Chinese file, both files are written (Section 5).

## 3.3 The All menu

*All* in the toolbar (Figure 13 **(1)**) opens a menu.

- **(2)** *Retranslate every paragraph* sends every paragraph again, without the cache. It is off while you edit the other language, since the right pane then holds the text of your file.
- **(3)** *Continue with the rest* translates only paragraphs without a translation or whose translation

failed, also while translation is paused. After an early swap, it also fills in paragraphs on the left still in the file's language.

- **(4)** *Read the document as Chinese, translate into English* (in a Chinese paper, the opposite) is for a document whose language was taken wrongly (Section 3.4).




Figure 13 (1) The All menu, (2) retranslate every paragraph, (3) continue with the rest, (4) change the document's language

## 3.4 A paper written in Chinese

Opening a file, BiWrite judges its language from the paragraphs of the text, with math, citations and LaTeX commands left out. When Chinese clearly prevails, you edit the Chinese on the left with the English translation on the right, and the status bar reads zh □ en (Figure 14). Then

- saving writes the Chinese on the left. The swap button lets you edit the English instead, and saving writes the Chinese composed from the translations back into the file,
- *Chinese* in the PDF tab compiles the Chinese file on disk, *English* compiles the English translation, and the export button reads *Export English .tex*….

In a mixed document the judgment can be wrong, and you can set it right. When the text on the left plainly is the other language, the swap button reads it as such at once, without waiting (Figure 15). Otherwise choose *Read the document as Chinese, translate into English* or the opposite in the *All* menu. The file stays as it is and is translated again from its language.

## 3.5 Save and Save As

*Save* (Figure 16 **(1)**, ⌘S) overwrites the file. *Save As* **(2)** (⌘⇧S) saves under a new name and suggests one not taken yet, for example `paper-2.tex`. The original stays as it was.

## 3.6 Glossary

*Glossary* in the toolbar opens the glossary (Figure 17). In the table **(1)**, set your preferred Chinese for a term or tick *Keep EN*. The buttons **(2)** import and export CSV files with the columns `term,translation` (UTF-8). After a change, only paragraphs that mention the term are translated again.

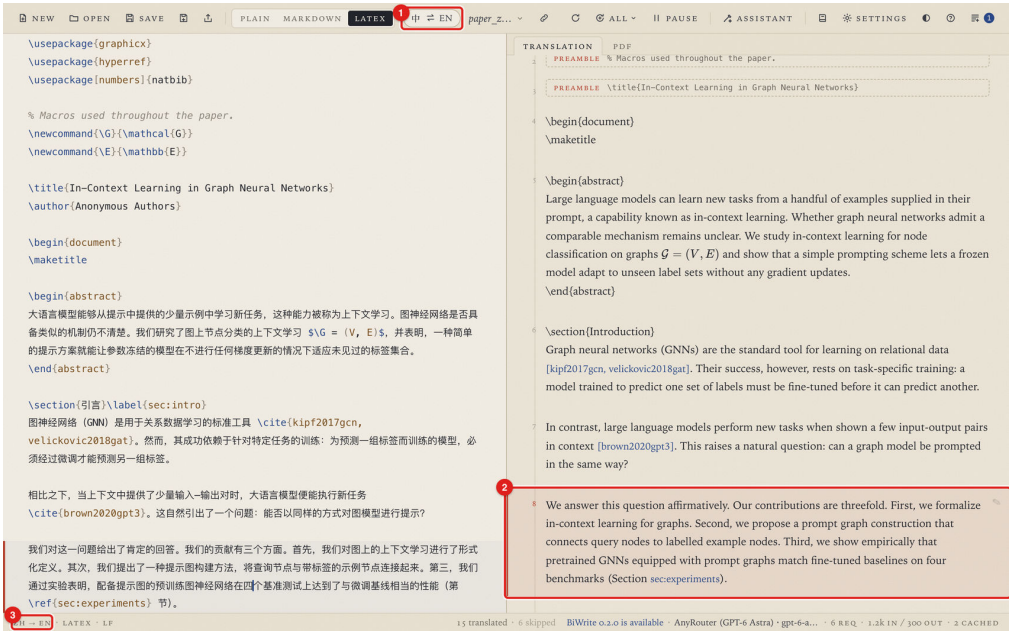



Figure 14 A Chinese file: (1) the button reads 中 ⇄ EN and is not swapped, (2) the English translation on the right, (3) zh □ en in the status bar

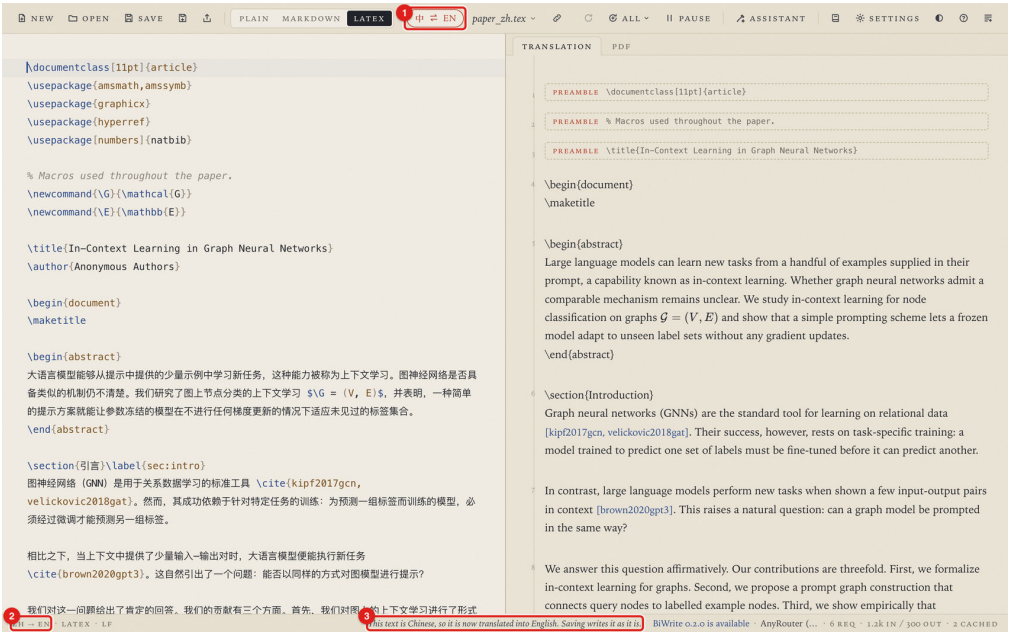



Figure 15 The swap button on a Chinese file taken for English: (1) the Chinese is now the source, (2) the status bar reads zh □ en, (3) the note

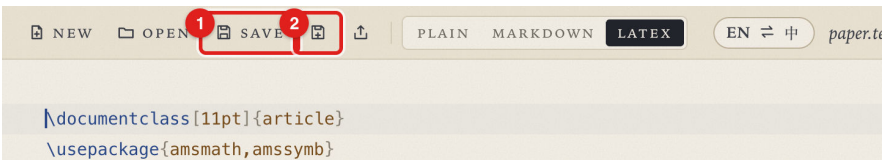



Figure 16 (1) Save, (2) Save As




Figure 17 The glossary: (1) terms and their Chinese, (2) import, export and save

# 4 LaTeX papers

## 4.1 Open a project

Open a `.tex` file with *Open*, or click *New* (Figure 18 **(1)**) and choose *Open LaTeX folder…* to open a project by its main file. A project of several files lists them in the title bar **(2)**, with the main file marked. The right pane of a LaTeX document has two tabs, *Translation* and *PDF* **(3)**.

BiWrite finds the main file from a `% !TEX root = …` comment, else from the `\input` and `\include` chains that reach the open file, however deeply nested. The engine comes from a `% !TEX program` comment, else the project's `latexmkrc` (`$pdf_mode`), else the packages of the preamble.




Figure 18 (1) New and Open LaTeX folder, (2) the files of the project, (3) the Translation and PDF tabs

## 4.2 Compile the English PDF

1. Click the *PDF* tab (Figure 19 **(1)**).
2. Click *Compile* **(2)** or press ⌘B (Ctrl+B on Windows). Unsaved edits are saved first, since the PDF comes from the files on disk.
3. The PDF appears. **(3)** shows the time taken and the numbers of errors and warnings, and the problem list **(4)** links each one to its line.

*Compile after saving a .tex file* is on by default, so every save compiles again. A document with errors still gets its PDF, with the errors listed.

## 4.3 From the PDF to the source

Click a sentence in the PDF (Figure 20 **(1)**) and the same sentence is selected in the source **(2)**. A small menu next to the click **(3)** offers *Polish*, *Edit…* and *Ask…* on that sentence. If the sentence is




Figure 19 (1) The PDF tab, (2) Compile, (3) the result, (4) the problem list

in another file of the project (such as `sections/04_method.tex`), BiWrite opens that file and selects it there, after asking when the open file has unsaved changes.

BiWrite checks the words under the click. Things drawn over the text, such as the line numbers in the margin of a submission or a running header, mislead SyncTeX, and then BiWrite looks the clicked words up in the open file and in the other files of the project.

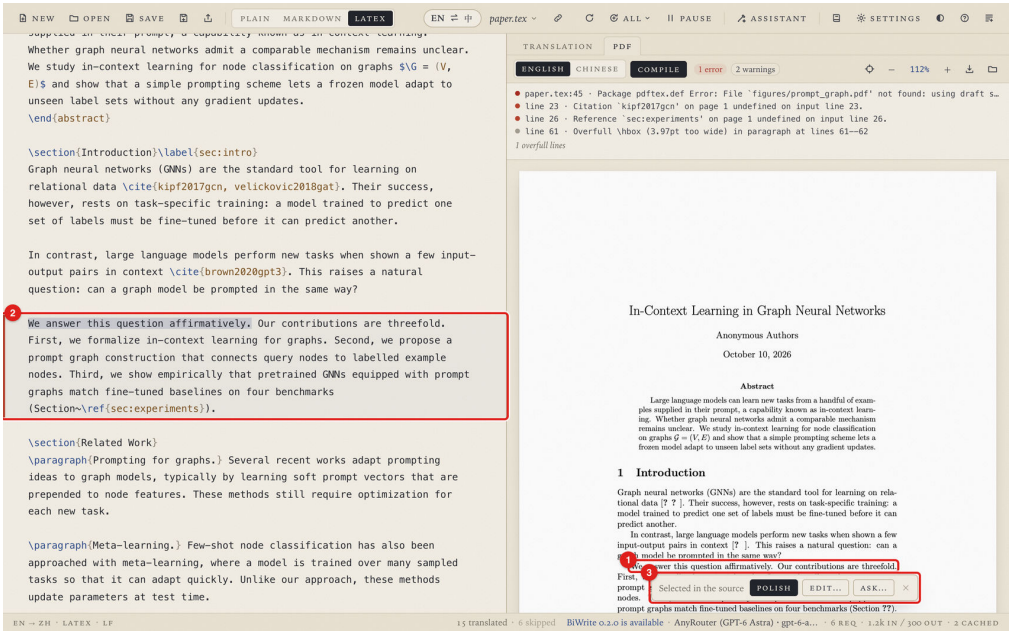



Figure 20 (1) A click on a sentence in the PDF, (2) the sentence selected in the source, (3) the menu

## 4.4 From the source to the PDF

Put the cursor in a paragraph (Figure 21 **(1)**) and click the locate button above the PDF **(2)**, or press ⌘⌥J (Ctrl+Alt+J). The paragraph is highlighted in the PDF for a few seconds **(3)**.

## 4.5 Compile the Chinese PDF after translation

The Chinese PDF is the translation set as a document, compiled with XeLaTeX with Chinese font support (the ctex package) added.

1. Open the English paper and wait until every paragraph is translated. The status bar counts translated, in-flight and pending paragraphs.
2. In the PDF tab, click *Chinese* (Figure 22 **(1)**). The first switch compiles it, and *Compile* **(2)** builds it again later.

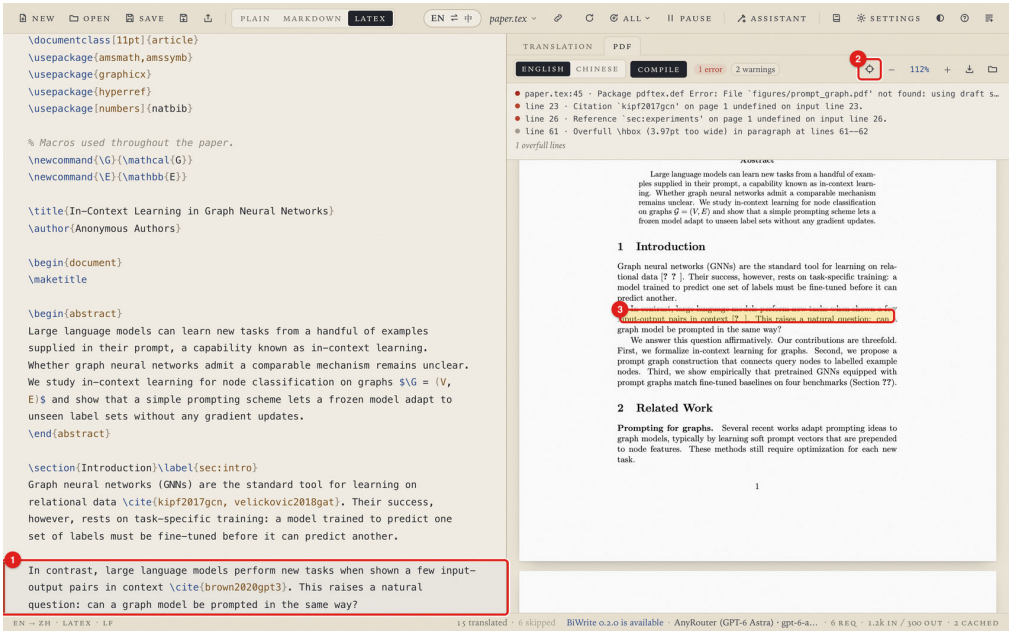


Figure 21 (1) The paragraph at the cursor, (2) the locate button, (3) its place in the PDF

3. Paragraphs not translated yet are set in English, and a note above the PDF counts them **(3)**.

The Chinese PDF and its build files are kept in the hidden folder `.biwrite/zh/` of the paper's folder. None of your files are touched.

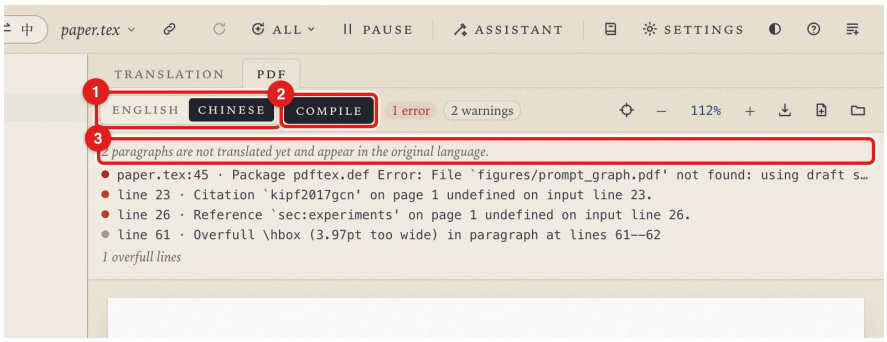



Figure 22 (1) Show the Chinese PDF, (2) Compile, (3) paragraphs not translated yet

## 4.6 Export the Chinese PDF and .tex

On the Chinese PDF (Figure 23):

- **(1)** *Save PDF as…* saves a copy of the Chinese PDF, named `paper.zh.pdf` by default. The same button saves the English PDF.
- **(2)** *Export Chinese .tex…* saves a LaTeX file that compiles on its own with XeLaTeX, named `paper_zh.tex` by default. A Chinese paper exports its English version here instead, with *Export English .tex…* on the English PDF.
- **(3)** *Show in folder* shows the PDF in Finder or Explorer.

## 4.7 Paper templates

*New* opens *Papers and templates* (Figure 24).

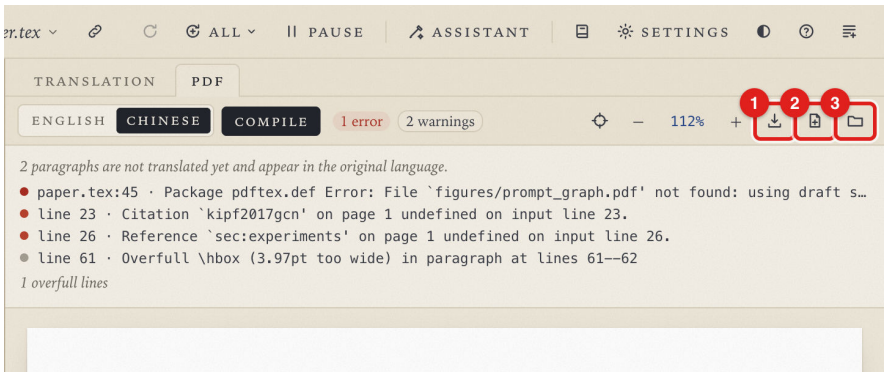



Figure 23 (1) Save PDF as, (2) Export Chinese .tex, (3) Show in folder

- The built-in templates are VLDB (PVLDB volume 20, 2027), ICLR 2027 and IEEE TII. The first two use the official template files, and TII comes with a short skeleton paper.
- **(1)** *New paper…* copies a template into a folder you name and opens the main file. The PDF tab compiles it.
- **(2)** *Open LaTeX folder…* opens an existing paper project.
- **(3)** *Import folder…* and *Import .zip…* keep an existing project as a template of your own, listed under *My templates*.
- **(4)** *Export this project…* zips the open project with a template manifest, for others to import.




Figure 24 (1) A new paper from a template, (2) Open LaTeX folder, (3) import a template, (4) export the project

## 5 An existing Chinese version

If you keep a Chinese version by hand (`paper_zh.tex` next to `paper.tex`), BiWrite uses it instead of translating. Once the two files are paired, editing the English updates the matching paragraph of the Chinese file, and editing the Chinese updates the English file.

### 5.1 Automatic pairing

Opening an English file looks for its Chinese counterpart by name and pairs them when their paragraphs match:

- a suffix in the same folder: `paper.tex` with `paper_zh.tex`, or `.zh`, `-zh`, `_cn`,

- parallel language folders: `sections_en/04_method.tex` with `sections_zh/04_method.tex`, `en/` with `zh/`.

## 5.2 Pair by hand

1. Open and save the English file.
2. Click *Pair* (Figure 25 **(1)**, the link icon) next to the file name in the title bar.
3. Choose the Chinese file. BiWrite aligns the two files paragraph by paragraph. The title bar shows the Chinese file (Figure 26 **(1)**), the status bar counts the paired paragraphs **(2)**, and the right pane shows your Chinese **(3)**.

Paragraphs pair by kind (paragraph, heading, caption), shared citation keys, references, labels, math, numbers and names, and length. A figure placed elsewhere in the Chinese file still finds its partner. Paragraphs without a partner are translated, but their translation is not written into the Chinese file.




Figure 25 (1) Pair

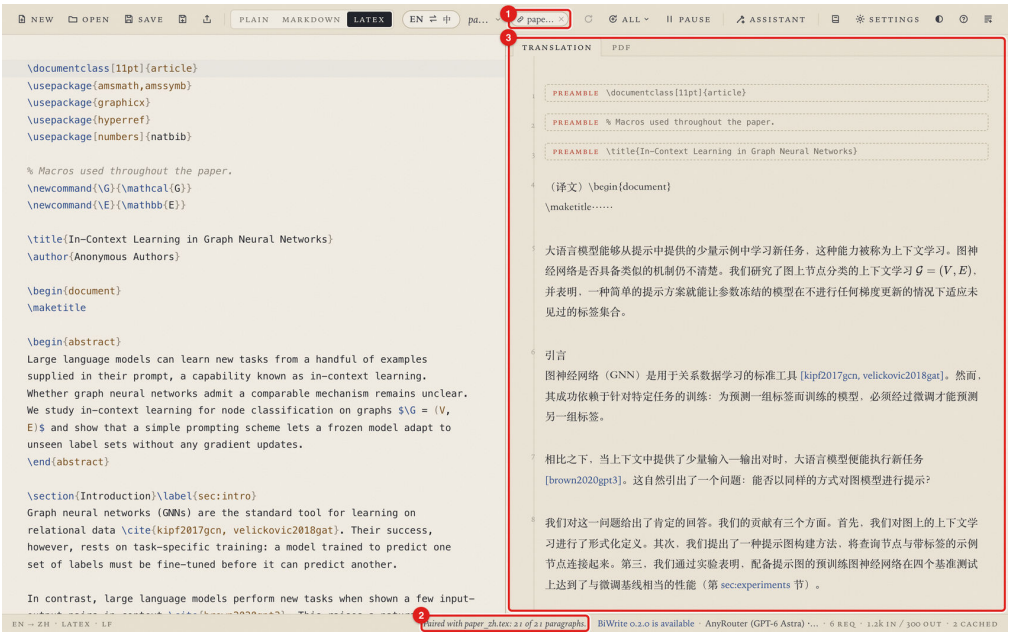



Figure 26 After pairing: (1) the Chinese file, (2) the paired paragraphs, (3) your Chinese on the right

## 5.3 Editing and saving a pair

- The right pane shows your Chinese, with no requests.
- Edit an English paragraph, and only its Chinese is revised, from your Chinese.
- You can also edit the Chinese directly. The pencil on a paragraph on the right (Figure 27 **(1)**) opens *Via translation* in the writing assistant with the paragraph's Chinese in the box **(2)**. Rewrite it and click *Run* **(3)**: the model revises the English as little as possible to say the same.

The card shows the change to the English (Figure 28 **(1)**), and *Accept* **(2)** applies it, with your sentence as the paragraph's Chinese.

- Saving writes both files. Only changed paragraphs are replaced in the Chinese file, and the rest stays byte for byte. If a translation is still on its way, the English is saved at once and the Chinese file follows when the translation arrives.
- After the swap, you edit your Chinese file on the left, and changes go back into the English file the same way.
- *Save As* (Figure 29 **(1)**) saves both files under the new name, for example `paper-2.tex` and `paper_zh-2.tex`. The original two stay as they were.




Figure 27 (1) The pencil, (2) the paragraph's Chinese to rewrite, (3) Run




Figure 28 (1) The change to the English, (2) Accept

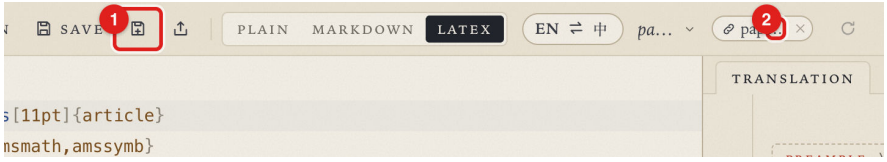



Figure 29 (1) Save As, (2) unpair

### 5.4 PDFs of a pair, and unpairing

With a pair, *Chinese* in the PDF tab compiles your own Chinese main file (`paper_zh.tex`), not a document assembled from the translation. A click in the Chinese PDF selects the matching English paragraph. The cross on the Chinese file in the title bar (Figure 29 **(2)**) ends the pair, and saving no longer writes the Chinese file.

## 6 Writing assistant

### 6.1 Open it and pick a task

*Assistant* in the toolbar or ⌘J opens the writing assistant on the right (Figure 30). It works on the selection, or on the paragraph at the cursor **(2)**.

1. Pick a task at **(1)**.

   **Polish**
   Applies the writing rules: it removes AI-style phrasing and dashes or semicolons that splice sentences, and keeps claims, numbers, citations and formulas. ⌘⇧P polishes the paragraph at the cursor.

   **Edit**
   Follows your instruction, for example *shorter, state the number of benchmarks*.

   **Ask**
   Answers a question about the selection or the whole paper, with follow-up questions.

   **Write**
   Writes new paragraphs as you ask, inserted after the paragraph, in the manner of a reference paper when you add one (Section 2, step 3).

   **Figure**
   Writes a LaTeX figure or table from a description, inserted after the paragraph with the packages it needs (Section 2, step 5).

   **Via translation**
   Fills the box with the paragraph's translation. Rewrite it and run, and the paragraph is revised to say the same (Section 5).

2. Write the instruction or question at **(3)** (polishing needs none).
3. **(4)** *Context* sends the text alone, with its neighbouring paragraphs, or with the whole paper.
4. **(5)** *References* takes passages whose style to follow (+ *Text*), whole papers as `.tex`, `.md` or `.txt` files (+ *File*), or images such as a figure whose style to follow (+ *Image*).
5. Click *Run* **(6)**. The job runs in the background while you keep writing.

### 6.2 Decide on the result

The finished card (Figure 31) shows the changes as a diff **(1)** (deleted words struck out, new ones highlighted), the translation of the revision **(2)** and the reasons for the changes in Chinese and English **(3)**. *Accept* **(4)** writes the revision where the text is now, even after you kept typing. If the target itself was edited meanwhile, the card says so, and *Re-apply with the model* brings the revision onto the current text. *Again* runs the job once more, *Discard* drops it, and ⌘Z undoes an accepted revision.

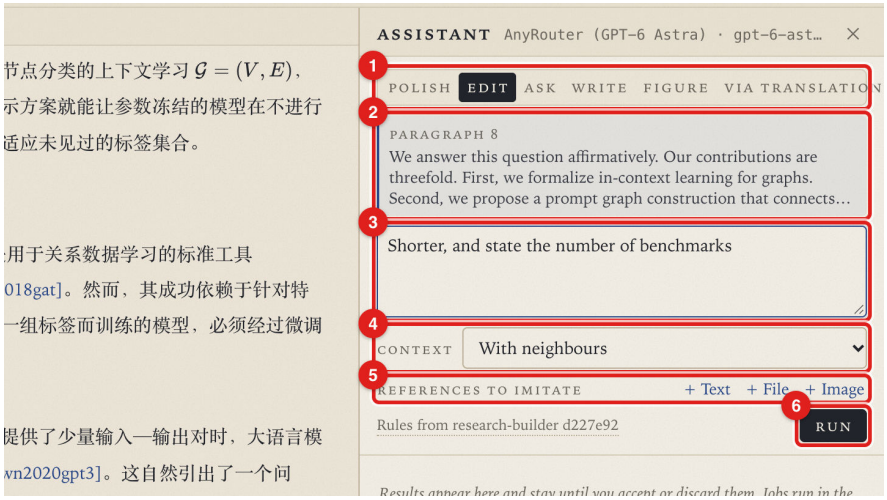



Figure 30 (1) Task, (2) target, (3) instruction, (4) context, (5) references, (6) Run

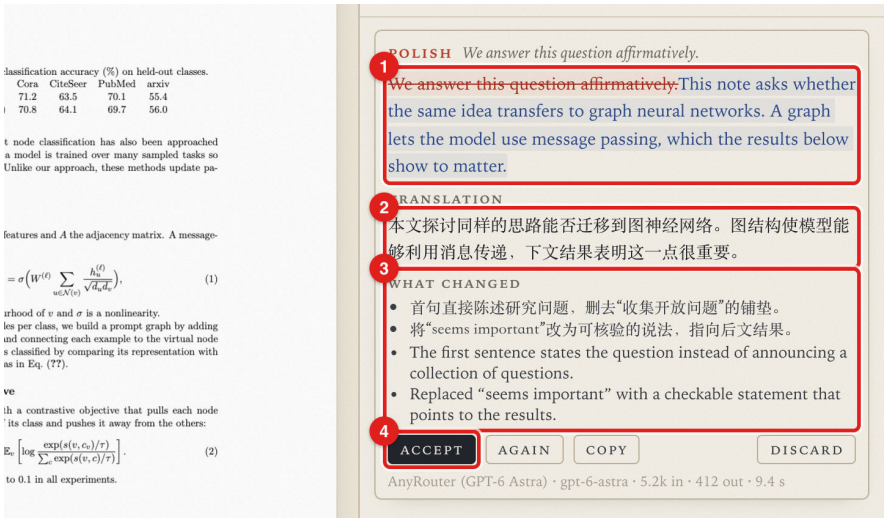



Figure 31 (1) The diff, (2) the translation, (3) the reasons, (4) Accept

## 6.3 Questions

An answer appears in the card (Figure 32 **(1)**), with a box for a follow-up question **(2)**. Writing new paragraphs and figures are shown in Section 2, steps 3 and 5.

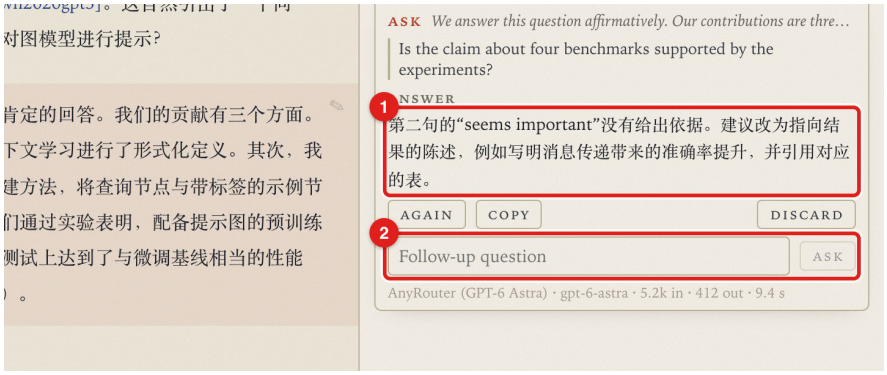



Figure 32 (1) The answer, (2) a follow-up question

### 6.4 The writing skill

The rules come from research-builder, an open-source writing skill under the MIT license. BiWrite ships a copy. The *Writing skill* section of Settings (Figure 33) can **(1)** update it from GitHub, **(2)** use a skill folder of your own, and **(3)** open the skill's repository.

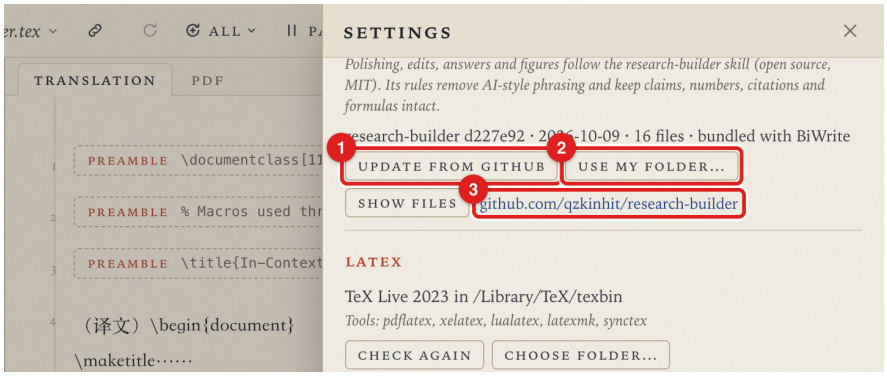



Figure 33 (1) Update from GitHub, (2) use your own folder, (3) the skill's repository

## 7 Request log

The log button at the right of the toolbar (⌘⇧L) opens the request log (Figure 34). **(1)** filters the records, pauses recording, clears the log and opens the log file. Each row of the table **(2)** is one request, with the model, reasoning effort and service tier that were sent and the ones the server declared, marked where they differ. A click on a row shows its details (Figure 35 **(1)**): the HTTP status, time to the first token, duration, input and output tokens (cached and reasoning tokens included), the key used (its name and last characters), and notes on retries and failover. The log never holds document text or keys.

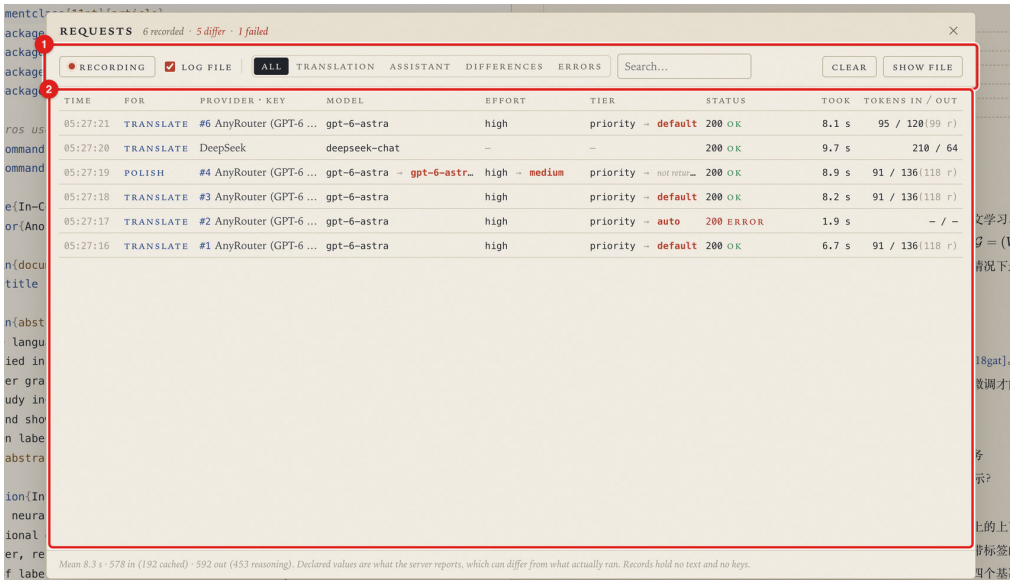



Figure 34 (1) Filters and actions, (2) the requests

## 8 Updates

In Settings, under *Updates*, *Show versions* (Figure 36 **(1)**) lists every release on GitHub with its notes **(2)**, older versions and pre-releases (branch builds) included. Pick a version and click *Install* **(3)**.




Figure 35 (1) The details of one request

BiWrite downloads the installer and checks it against the SHA-256 that GitHub reports. On macOS, *Restart now* (Figure 37 **(1)**) starts the new version. On Windows, the installer runs after BiWrite quits. At startup, BiWrite looks for a newer release and says so in the status bar, which can be turned off.

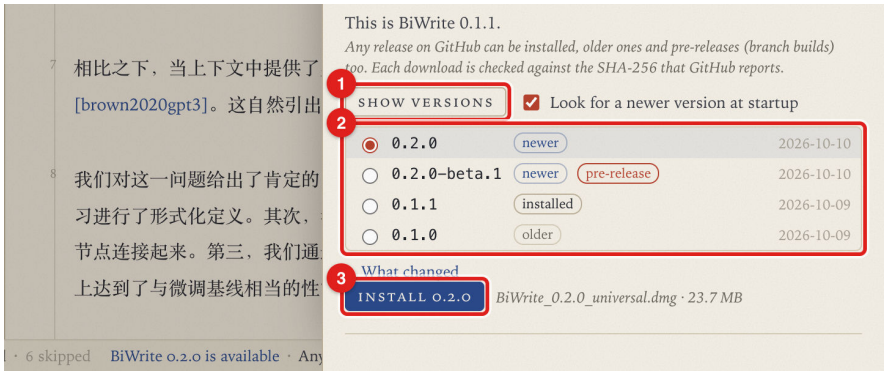



Figure 36 (1) Show versions, (2) the releases, (3) Install




Figure 37 (1) Restart after the installation

# 9 Install

## 9.1 macOS

Download `BiWrite_<version>_universal.dmg` from the Releases page (`https://github.com/zzccppp/biwrite/releases`). It runs on Apple silicon and Intel Macs with macOS 12 or later. Open the disk image and drag BiWrite into *Applications*.

The app is not signed with an Apple developer certificate, so macOS blocks it the first time. To allow it:

1. Double-click BiWrite. macOS says it cannot verify the developer. Click *Done*.
2. Open *System Settings* and go to *Privacy & Security*.

3. Under *Security* near the bottom, macOS shows *"BiWrite" was blocked to protect your Mac*. Click *Open Anyway*.
4. Confirm with *Open Anyway* again and enter your password or use Touch ID.

After this, BiWrite opens normally. In Terminal, this command has the same effect:

```
xattr -dr com.apple.quarantine /Applications/BiWrite.app
```

### 9.2 Windows

Download `BiWrite_<version>_x64-setup.exe` for 64-bit Windows 10 or 11. The installer has no trusted code signature either. When SmartScreen shows *Windows protected your PC*, click *More info*, then *Run anyway*, and follow the installer.

### 9.3 A TeX distribution

LaTeX papers need a TeX distribution: MacTeX on macOS, TeX Live or MiKTeX on Windows. BiWrite looks for it in the usual places and shows what it found in the *LaTeX* section of Settings (Figure 38 **(1)**). For a distribution somewhere else, *Choose folder…* **(2)** picks the folder that holds `pdflatex`. **(3)** turns compiling after every save on or off.

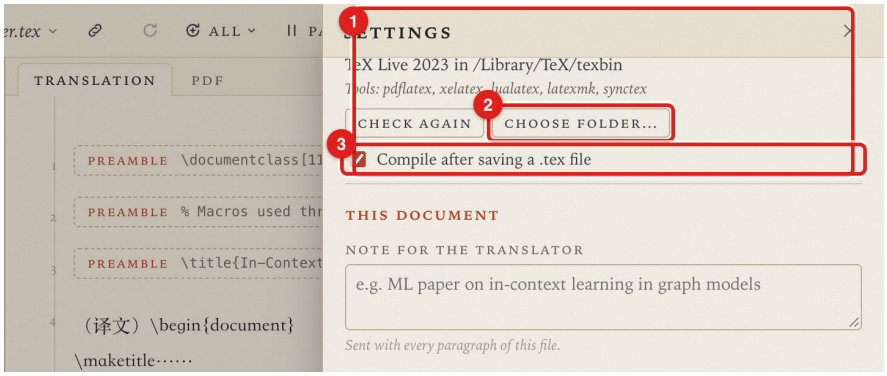



Figure 38 (1) The TeX distribution found, (2) Choose folder, (3) compile after saving

## 10 Set up a model provider

### 10.1 Supported APIs

Translation and the writing assistant use a language model API. BiWrite supports:

- OpenAI-compatible Chat Completions APIs: OpenAI, DeepSeek, Qwen, Kimi, OpenRouter and local servers such as Ollama or LM Studio,
- the OpenAI Responses API, for reasoning models that take a reasoning effort and a service tier,
- the Anthropic Messages API.

The built-in *Mock* provider works offline and returns each paragraph reversed, for trying the interface without a key.

### 10.2 Add a provider

1. Open *Settings* (⌘,). **(1)** lists the providers, and the radio button selects the one that translates (Figure 39).

2. Choose + *Add provider…* at **(2)** and pick a preset.
3. In the provider's form (Figure 40), check the base URL, the model and the API type **(1)**, and choose the reasoning effort and service tier **(2)** where the API takes them.
4. Paste the key and click *Add provider*. Keys are kept in the system keychain (the Credential Manager on Windows) and are never shown again. *Manage keys…* **(3)** manages them later.
5. *Test* **(4)** translates one sentence.

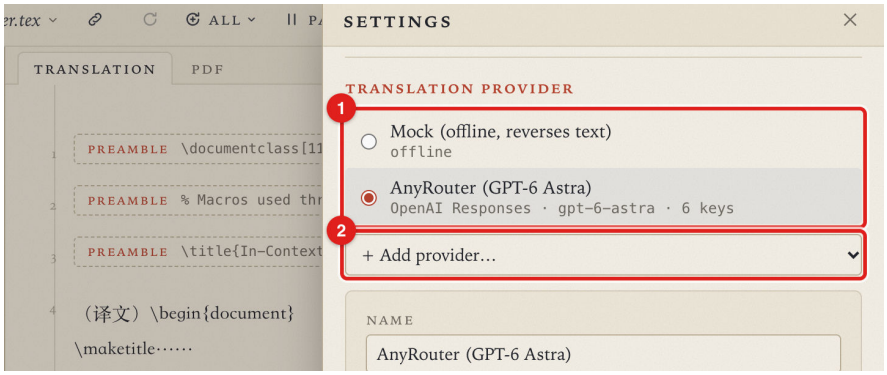



Figure 39 (1) The providers, (2) add a provider




Figure 40 The provider's form: (1) API type and temperature, (2) reasoning effort and service tier, (3) Manage keys, (4) Test

### 10.3 A third-party API: AnyRouter as an example

AnyRouter is a third-party service compatible with the OpenAI Responses API. The *AnyRouter (GPT-6 Astra)* preset sets the Responses API, the address `https://anyrouter.top/v1`, the model `gpt-6-astra`, high reasoning effort, the *Priority* service tier, two requests per key and ten retries. Paste one or more keys and save. Other third-party APIs are set up the same way, with the address, model and API type from the provider's documentation. The request log shows the reasoning effort and service tier the server reports back, so you can check that a third-party service honours them.

## 10.4 Key pools

A provider can hold several keys, a pool. *Manage keys…* in the provider's form opens it (Figure 41).

- **(1)** lists the keys. Give each a name. The table shows only the last characters of a key, its state and its requests in flight. Each request goes to the least busy ready key. A key hitting a rate limit cools down and the request moves to another key, and a rejected key is set aside.
- **(2)** sets the requests per key and the retries. AnyRouter allows two conversations per account, hence 2.
- **(3)** takes new keys: one per line, or a name on one line and its key on the next, or *name: key*. *Add keys* adds them to the pool.
- **(4)** *Import from file…* and *Export to file…*: the export saves the names and keys to a text file to move them to another computer or share them with a colleague, and the import adds the keys of such a file. The keys go from the keychain to the file without passing through the window. The file holds the keys in plain text, so share it only with people you trust.




Figure 41 The key pool: (1) the keys, (2) requests per key and retries, (3) new keys, (4) import from and export to a file

**Squeezing in.** A relay such as AnyRouter, when it is full, answers only requests with keys that got through a while ago. Requests with the other keys get HTTP 500, and their paragraphs fail with provider server error (HTTP 500). For AnyRouter, or a pool of two keys or more, the row **(2)** of the key pool ends with *Squeeze in*. BiWrite then sends a very short translation with every key that has not got through, all at once, and tries each failed key again every few seconds until every usable key got through. A key that is invalid or out of quota, or whose request cannot succeed (a misspelt model, for example), is not tried again. *Stop squeezing* stops it at any time, and so do adding or removing keys and changing the provider's address, model or API type. Requests the relay turns away cost nothing. A key that got through shows *through* with the time it got through, and hovering shows when it was last answered. A key not answered for more than five minutes may have lost its place, so its state changes colour and a running squeeze tries it again. The squeeze runs in the background, so the key pool can be closed and another provider can translate meanwhile. The status bar shows the progress. When a key gets through, a reminder at the bottom right shows how long ago a key of this provider was last answered. It stays until *Got it*, or until no key of this

provider counts as through. While another provider is in use, *Switch to it* on the reminder goes back to this one.

Once the relay turned away keys that had not got through twice within five minutes, translations and the writing assistant send their requests to the keys that got through first, and wait for one of them when they are all busy. Such a key that answers HTTP 500 now and then is not paused and is asked again soon. After six failures in a row, after a failure that follows five minutes without an answer, or when the key is refused, it no longer goes first. Otherwise the keys take turns. For providers other than AnyRouter this applies only after *Squeeze in* was used.

## 10.5 Separate models for translation and writing

The *Writing assistant* section of Settings (Figure 42 **(1)**) picks the model for polishing, edits and questions. A cheap, fast model can translate while a stronger one writes. *Same as translation* uses one provider for both.




Figure 42 (1) The model of the writing assistant

## 10.6 Request pacing

The *Requests* section (Figure 43) controls how many requests reach the provider.

- **(1)** Wait after an edit: 5 seconds by default, 0 to 60. An edited paragraph is sent for translation once it has stayed unchanged this long, so editing it does not send a request at every pause. At 0 it is sent at once.
- **(2)** Paragraphs per request: 1 by default. With 2 to 8, several new paragraphs share one request, which suits providers with rate limits. Edited paragraphs are always sent alone, so their translation is revised minimally.
- **(3)** Parallel requests: the number of requests at once.
- **(4)** With *Use the whole key pool*, it is the number of keys times the requests per key.

## 10.7 Interface language

*Language* at the top of Settings (Figure 44 **(1)**) switches the interface between English and Chinese. Translation directions do not change.

## 10.8 Closing the window and the tray

In the *Window* section of Settings (Figure 45 **(1)**), *Closing the window hides BiWrite in the tray* is on by default. Closed, the window hides and BiWrite keeps running with the open document, and

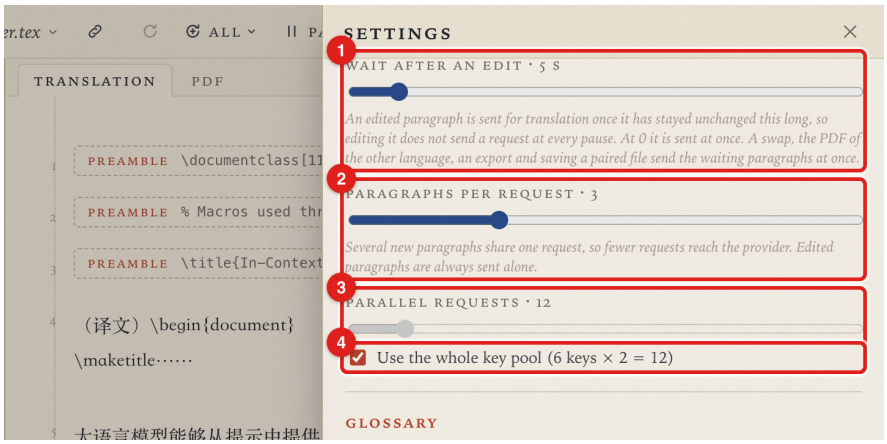



Figure 43 (1) Wait after an edit, (2) paragraphs per request, (3) parallel requests, (4) follow the key pool




Figure 44 (1) The interface language

translations and assistant jobs go on. The BiWrite icon in the menu bar (macOS) or the notification area (Windows) offers *Show BiWrite* and *Quit BiWrite*, and on macOS the Dock icon brings the window back too. Quitting with unsaved changes asks first. Turned off, closing the window quits BiWrite.

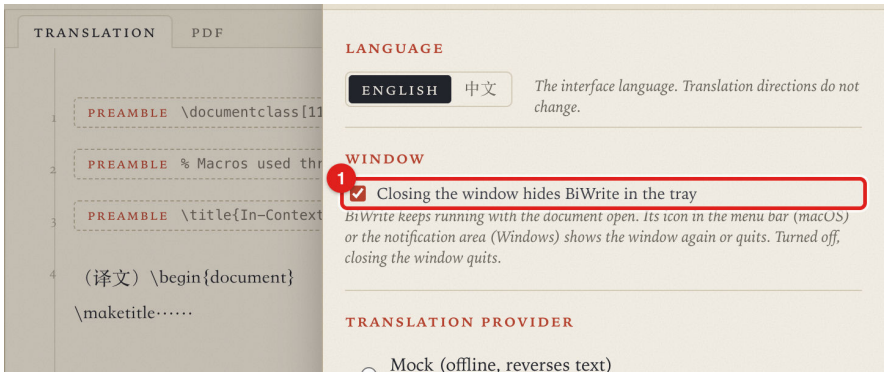



Figure 45 (1) Closing the window hides BiWrite in the tray

# 11 Keyboard shortcuts

| macOS (Ctrl for ⌘and Alt for ⌥on Windows) | Action |
|---|---|
| ⌘O | Open a file |
| ⌘S / ⌘⇧S | Save / Save As |
| ⌘B | Compile the PDF on show |
| ⌘⌥J | Find the cursor in the PDF |
| ⌘J | Open or close the writing assistant |
| ⌘⇧P | Polish the paragraph at the cursor |
| ⌘K | Focus the assistant's input |
| ⌘, | Settings |
| ⌘⇧L | Request log |
| ⌘F | Find and replace in the source |
| ⌘Z | Undo (accepted revisions too) |

# 12 Questions

**No TeX distribution was found.** Install MacTeX, TeX Live or MiKTeX and click *Check again* in the PDF tab. For a distribution in another place, choose its folder in the *LaTeX* section of Settings.

**The build failed or the PDF is old.** Read the problem list above the PDF and click an error to reach its line. For a failure without a log, *Console output* shows the tools' output. A build stops after five minutes.

**A click in the PDF does nothing.** Every PDF that BiWrite builds has SyncTeX data. A click on an image or empty space shows that it has no source line.

**A click says the PDF is being built again.** After pairing or unpairing a translation, or swapping languages without a pair, the PDF built before no longer matches the text on the left, and the PDF tab says it is from before the change. A click or the cursor then builds that PDF again first, without saving any file. Click once more when it is ready.

**The swap button keeps waiting.** Some paragraphs are still being translated or failed. Click *Retry* on a failed paragraph or choose *Continue with the rest* in the *All* menu, or click the swap button again to swap at once.

**A Chinese file shows Chinese on the right.** BiWrite took the text for English, which happens with mixed documents. Choose *Read the document as Chinese, translate into English* in the *All* menu. The file stays as it is.

**An inserted figure does not compile.** Check the packages named on the card and load them in the main file's preamble. In the main file itself, inserting adds them.

**Where did BiWrite go after closing the window?** It hides in the tray and keeps running. The BiWrite icon in the menu bar or notification area brings the window back, and the *Window* section of Settings turns this off (Section 10.8).

**A third-party API often hits rate limits.** Add several keys to the pool, set *Requests per key* to what the provider allows, raise *Paragraphs per request* and *Retries*. The request log shows each rate limit and each switch to another key. When paragraphs keep failing with provider server error (HTTP 500), use *Squeeze in* in the key pool.

**The Chinese file does not pair.** Files pair only when at least half of their paragraphs match. Check that the two files are the same document in two languages.

**Where is the data kept?** Translations are cached in `cache.sqlite3` in the app data folder. The *Translation cache* section of Settings shows its size and clears it. Keys are kept only in the system keychain.